

AMR Individual Report

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Contributions to the project:

I conceptualized the initial idea for our automatic metamorphic relation discovery project, pitched it to the team during our early meetings, and proactively contacted the professor to ensure our approach was sound and approved before we committed to it. I also took the lead on our midterm delivery, creating and co-presenting the midterm presentation with Ada. For the final presentation, we split the delivery again; I designed the slides, and presented the introduction and methodology, while Ada covered the tool's evaluation.

On the technical side, I was responsible for modules related to the discovery of the MRs and their parameters, mainly:

- `amr/bounds.py`:

I built the profile-to-bounds translation module, which takes the high-level properties provided by the LLM and converts them into mathematical search boundaries.

- `amr/pso.py`

The Particle Swarm Optimization algorithm. This served as our standard optimization engine, running within the defined bounds to find the optimal coefficients for our relation template without overcomplicating the search dynamics.

- `amr/mr.py`

The most complex part of my contribution, i.e. the fitness function and validation logic.

Because a naive search would easily find mathematically useless relations, I made a fitness function that evaluates the Mean Squared Error (MSE) of the relation over sample inputs while enforcing strict penalties.

The validation step acts as our primary safeguard against overfitting by re-evaluating each candidate relation on a fresh set of inputs drawn from a different seed. A candidate is only accepted if its maximum residual remains below our tolerance on these unseen points, and if it continues to pass all non-triviality checks.

Additional info:

During the presentation I wasn't able to correctly address the professor's question about the "Constant output" penalty. I want to answer it directly here:

By penalizing values where the scaling factor drops near zero, we prevent the search from evaluating $P(x^2)$ as essentially $P(b)$, which is a constant. Without this penalty, the search might cheat by finding a static offset rather than a true dynamic relationship between two varying points in the domain.

Peer evaluation:

Yeskendir Assankul 9/10

Ada Jacyna 9/10

Jakub Skibicki 9/10

I think our team coordinated very well, we had no quarrels and everyone delivered their part on time.

Yeskendir came up with a good idea to introduce a twist to our project by using an LLM for the discovery optimization.

Ada helped me on stage by presenting the most data and knowledge heavy slides.

And Kuba quickly started to coordinate our work environment by creating the repository.